Group II Row IV: the running coupling and the $d^3N^y_{\xi\mathrm{LO}}$ term

Notation: $X \equiv y' - y$, $c_0 = 2e^{-\gamma_E}$, $\hat\mu^2 \equiv \mu^2_{\overline{MS}} e^{2\gamma_E}/4 = \mu^2_{\overline{MS}}/c_0^2$, $a = k^+/\vee$, $\frac{g^2}{16\pi^2} = \frac{\alpha_s}{4\pi}$. LO$[f]$ is Eq. (5.15) with the weight $f(X)$ *inside* $\int_{x,y,x',y'}$ (likewise $\frac{d^3N_{\xi\mathrm{LO}}}{d^3k}[f]$, $\frac{d^3N^y_{\xi\mathrm{LO}}}{d^3k}[f]$).

1 Running coupling: it matches

Step 1: the ϵ-bracket is $\ln(X^2\hat\mu^2)$.

$$2\gamma_E + \log\frac{X^2\mu^2_{\overline{MS}}}{4} = \log\frac{X^2\mu^2_{\overline{MS}}e^{2\gamma_E}}{4} = \ln\left(X^2\hat\mu^2\right). \tag{1.1}$$

Step 2: remove the pole. The RC line of Eq. (5.12) is the bare term. LO is $\propto g^2$, and the $\overline{MS}$ coupling counterterm adds $-\frac{\alpha_s\beta_0}{4\pi}\frac{2}{\epsilon}$LO:

$$\frac{g^2\beta_0}{16\pi^2}\mathrm{LO}\Big[\frac{2}{\epsilon} + \ln(X^2\hat\mu^2)\Big] - \frac{\alpha_s\beta_0}{4\pi}\frac{2}{\epsilon}\mathrm{LO} = \frac{\alpha_s\beta_0}{4\pi}\,\mathrm{LO}\big[\ln(X^2\hat\mu^2)\big]. \tag{1.2}$$

Step 3: compare. LO $= \alpha_s \times (\dots)$ with one power of α_s ($\frac{g^2}{2\pi^2} = \frac{2\alpha_s}{\pi}$). So

$$\mathrm{LO}\big|_{\alpha_s} + (1.2) = \frac{1}{(2\pi)^3}\frac{2}{\pi k^+}\int_{x,y,x',y'} e^{-ik\cdot X}\Big[\alpha_s + \frac{\alpha_s^2\beta_0}{4\pi}\ln\left(X^2\hat\mu^2\right)\Big]\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}[\dots]\rho\rho = \mathrm{LO}\big|_{\alpha_s\to\alpha_s(X^2)} + O(\alpha_s^3), \tag{1.3}$$

with exactly your advisor's

$$\alpha_s(X^2) = \alpha_s + \frac{\alpha_s^2\beta_0}{4\pi}\ln\left(X^2\hat\mu^2\right) + O(\alpha_s^3), \qquad X = y' - y \tag{1.4}$$

(`checks.py`: difference 0, sympy). It matches if three points are respected:

- **The $\frac{2}{\epsilon}$ has to be removed.** The bare line still has it; the coupling counterterm removes it.
- **$X^2 = (y'-y)^2$ sits inside the integral.** y, y' are where the produced gluon crosses the target in the amplitude and in the conjugate amplitude. The coupling runs with this separation, $|X| \sim 1/|\mathbf{k}|$.
- **The DGLAP line must use the full $\mathcal{P}_{gg}$**, including $\frac{\beta_0}{2}\delta(1-\xi)$. If instead DGLAP is written with the real part $\mathcal{P}_{gg} - \frac{\beta_0}{2}\delta(1-\xi)$ only, there is no separate RC line: the δ part at μ_F plus (1.2) at μ_R give $\frac{\alpha_s\beta_0}{4\pi}\ln\frac{\mu_R^2}{\mu_F^2}$LO (checked). The two ways of writing it are identical.

2 The $d^3N^y_{\xi\mathrm{LO}}$ term: your advisor is right, it combines with H_4

Step 1: why it looks asymmetric. In Eq. (5.14) the kernel and the Wilson line are taken at y, and y' is left only in the phase and in the log. As written (log outside) it is even zero: $\int_{y'} e^{-ik\cdot(y'-y)/\xi} = (2\pi\xi)^2\delta^2(\mathbf{k})$. With the log inside, do the y' integral. The μ and the constant multiply $\delta^2(\mathbf{k})$ and drop; the rest is

$$\int d^2y'\, e^{-ik\cdot(y'-y)/\xi}\log(y'-y)^2 = -\frac{4\pi\xi^2}{\mathbf{k}^2} \qquad \text{(checked)}, \tag{2.1}$$

so

$$\frac{d^3N^y_{\xi\mathrm{LO}}}{d^3k}\Big[\log\frac{X^2\mu^2}{4}\Big] = -\frac{4\pi\xi^2}{\mathbf{k}^2}\,\mathrm{LO}^y, \tag{2.2}$$

$$\mathrm{LO}^y \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{x,x',z}\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}\big[U^{e'c}(z) - U^{e'c}(x')\big]\big[U^{ec}(z) - U^{ce}(x)\big]\rho^{e'}(x')\rho^e(x) \tag{2.3}$$

($y \to z$). This is symmetric under $(x, \text{amplitude}) \leftrightarrow (x', \text{conjugate})$: both gluon points sit at the same z.

Step 2: the H_4 term after the w, w' integrals. With $\int d^2w\, e^{ik\cdot w}\frac{(w-z)^i}{(w-z)^2} = \frac{2\pi i k^i}{\mathbf{k}^2}e^{ik\cdot z}$ (checked), the w, w' integrals give $\frac{4\pi^2}{\mathbf{k}^2}$, and

$$H_4 \equiv \frac{1}{(2\pi)^3}\frac{g^4 N_c}{8\pi^5 k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2}\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}[\dots]\rho\rho = \frac{g^2 N_c}{\pi\mathbf{k}^2}\,\mathrm{LO}^y. \tag{2.4}$$

So the two terms have the same structure, as your advisor says.

Step 3: add them. The d^3N^y term of Eq. (5.12):

$$-\frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\Big(-2N_c\frac{1-\xi}{\xi}\Big)\Big(-\frac{4\pi\xi^2}{\mathbf{k}^2}\Big)\mathrm{LO}^y = -\frac{g^2 N_c}{\pi\mathbf{k}^2}\mathrm{LO}^y\int_a^1 d\xi\frac{1-\xi}{\xi} = -\Big[\log\frac{\vee}{k^+} - \Big(1-\frac{k^+}{\vee}\Big)\Big]H_4. \tag{2.5}$$

With the H_4 line of Eq. (5.12), the $(1-\frac{k^+}{\vee})$ parts cancel exactly:

$$-\Big[\log\frac{\vee}{k^+} - 1 + \frac{k^+}{\vee}\Big]H_4 - \Big(1-\frac{k^+}{\vee}\Big)H_4 = -\log\frac{\vee}{k^+}\,H_4. \tag{2.6}$$

Step 4: and with the BFKL line it cancels completely. The full Row IV also has $+\log\frac{\vee}{k^+}H_4$, which went into the BFKL list. So in the full Row IV the three H_4 structures cancel:

$$\Big(d^3N^y\text{ term}\Big) + \Big(-(1-a)H_4\Big) + \Big(+\log\tfrac{\vee}{k^+}H_4\Big) = 0. \tag{2.7}$$

Why it was there. The DGLAP term $-\frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\mathcal{P}_{gg}\frac{d^3N_{\xi\mathrm{LO}}}{d^3k}[\dots]$ contains $+[\log\frac{\vee}{k^+}-(1-a)]H_4$ at its lower end $\xi\to a$, where the phase forces $y'\to y$. The d^3N^y term is just that piece, written separately to show it. It is not a new term.

3 Group II Row IV, written symmetrically

$$\begin{aligned}\text{Row IV} = {}& \underbrace{\frac{g^2N_c}{4\pi^2}\log\frac{k^+}{\Lambda}\,\mathrm{LO}\Big[-\frac{2}{\epsilon}-\ln(X^2\hat\mu^2)\Big]}_{\text{JIMWLK}} \\ & \underbrace{-\frac{g^2}{8\pi^2}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\,\frac{d^3N_{\xi\mathrm{LO}}}{d^3k}\Big[\frac{2}{\epsilon}+\ln(X^2\hat\mu^2)\Big]}_{\text{DGLAP}} + \underbrace{\frac{g^2\beta_0}{16\pi^2}\,\mathrm{LO}\Big[\frac{2}{\epsilon}+\ln(X^2\hat\mu^2)\Big]}_{\text{RC}} \\ & + \underbrace{\frac{g^2N_c}{4\pi^2}\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\,\frac{d^3N^{\mathrm{lp}}_{\xi\mathrm{LO}}}{d^3k}}_{\text{finite}},\end{aligned} \tag{3.1}$$

with no d^3N^y term and no H_4.

- After the counterterms, DGLAP $\to -\frac{\alpha_s}{2\pi}\int\frac{d\xi}{\xi^2}\mathcal{P}_{gg}\frac{d^3N_{\xi\mathrm{LO}}}{d^3k}[\ln(X^2\mu_F^2/c_0^2)]$ and RC $\to \mathrm{LO}|_{\alpha_s\to\alpha_s(X^2)}$.
- **Bookkeeping.** If the BFKL list keeps $+\log\frac{\vee}{k^+}H_4$ (the H_4 piece of BFKL⊗LO, which sits inside the DGLAP term's lower end), Row IV must carry $-\log\frac{\vee}{k^+}H_4$, Eq. (2.6), to avoid counting it twice.

Checks (`checks.py` → `checks.json`)

RC after the counterterm $-\frac{\alpha_s\beta_0}{4\pi}\ln(X^2\hat\mu^2)$; $\hat\mu^2=\mu^2/c_0^2$	0; 0 (sympy)
δ-part of $\mathcal{P}_{gg}$ at μ_F + RC at μ_R	$\frac{\alpha_s\beta_0}{4\pi}\ln\frac{\mu_R^2}{\mu_F^2}$
$2\pi\int_0^\infty J_1(kr)e^{-\delta r^2}dr$ vs $\frac{2\pi}{k}$ $(k=1.3)$	4.83322 vs 4.83322
$2\pi\int rJ_0(qr)\log r^2 e^{-\delta r^2}dr \to -\frac{4\pi}{q^2}$ $(q=1.3)$	-7.489, $-7.453\to-7.436$
$H_4 = \frac{g^2N_c}{\pi\mathbf{k}^2}\mathrm{LO}^y$	exact
d^3N^y term $+(-(1-a)H_4)$ in units of H_4	$\log a = -\log\frac{\vee}{k^+}$